BT5110 Tutorial 3

Functional Dependencies

Your company, Apasaja Private Limited, is commissioned by Toko Kopi Luwak to design a relational schema for the management of their coffee beans, drinks, and cafes.

A coffee bean is identified by its unique brand name or a combination of its cultivar and region (since the same cultivar can be grown in different regions). For instance, we may have a coffee bean named “The Waterfall” which comes from a Tabi cultivar grown in Colombia.

A drink can be made using a particular coffee bean. The name of the drink is only unique for a particular coffee bean. This means we can have an “Espresso” made with “The Waterfall” or “La Bella” (which is a Pacamara cultivar grown in Guatemala). The price of the drink is also recorded.

A branch identified by its branch name may sell the drink. A drink may be sold by zero or more branches, and a branch may sell zero or more drinks. The address of the branch is also recorded. For each drink sold by a branch, we record the quantity sold to see which branch is the most profitable.

An abstract schema for this application is given below.

R = {A, B, C, D, E, F, G, H}

Σ = { {A} → {C, E}, {A, B} → {D}, {F} → {H}, {C, E} → {A}, {B, C, E} → {D},
{A, B, F} → {D, G}, {B, C, E, F} → {G} }

1. Functional Dependencies

(a) From the functional dependencies in Σ and the text description of the application, can you figure out the mapping of the attributes and the letters?

(b) Compute the attribute closures of the subsets of attributes of R with Σ in order to find the candidate keys of R with Σ.

(c) Find the prime attributes of R with Σ.

2. Minimal Cover

(a) Compute a minimal cover of R with Σ.

(b) Compute a canonical cover of R with Σ.